

Machine Learning and Model Theory

Stephen Drake, Tomislav Stojcic

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1 Introduction

Machine learning has become ubiquitous in everyday life, powering applications ranging from recommendation systems and navigation services to large language models. There are generally three different paradigms: unsupervised learning, reinforcement learning, and supervised learning. While the first two have shown massive potential and promise, supervised learning remains by far the most prominent today and will encompass our attention for this report.

The fundamental goal of supervised learning is seemingly simple: given an input x and an output y , we seek to find a function h (a hypothesis) such that $h(x) = y$. However, a core difficulty arises because we do not have access to the full population distribution of data. Instead, we must rely on a training sample $S = \{(x_1, y_1), \dots, (x_m, y_m)\}$, a finite sequence of observations.

This creates the problem of *generalization*: how can we guarantee that a function that performs well on a finite sample will also perform well on unseen data? Without a rigorous mathematical framework, a model might simply memorize the training set (overfitting) without learning the underlying rules. To address this, we will turn to Statistical Learning Theory, which provides the formal bounds required to measure a model's ability to learn.

Interestingly, while Vapnik and Chervonenkis were laying the groundwork for Statistical Learning Theory in the late 1960s and early 1970s, almost simultaneously, and in a seemingly completely different universe, Saharon Shelah was revolutionizing Model Theory with his work on stability and classification theory. As we will soon explore, these two areas turn out to share a profound structural connection.

1.1 The Learning Framework

Let X be a non-empty set and $Y = \{0, 1\}$. Let $\mathcal{H} \subseteq Y^X$ be a hypothesis class of binary functions. We will work in the **agnostic setting**, where we do not assume the existence of a perfect target function $f \in \mathcal{H}$. Instead, we assume that data is generated by an arbitrary, unknown distribution $\mathcal{D}$ over $X \times Y$.

Definition 1.1 (Risk). *The generalization error (or risk) of a hypothesis $h \in \mathcal{H}$*

with respect to a distribution $\mathcal{D}$ is the probability of misclassification:

$$L_{\mathcal{D}}(h) := \mathbb{P}_{(x,y) \sim \mathcal{D}}[h(x) \neq y].$$

Definition 1.2 (PAC Learnability). *A hypothesis class $\mathcal{H}$ is PAC learnable if there exists a learning algorithm $\mathcal{A}$ and a sample complexity function $m_{\mathcal{H}} : (0, 1)^2 \rightarrow \mathbb{N}$ such that for every $\epsilon, \delta \in (0, 1)$ and for every distribution $\mathcal{D}$ over $X \times Y$, if $\mathcal{A}$ receives a sample S of size $m \geq m_{\mathcal{H}}(\epsilon, \delta)$ drawn i.i.d. from $\mathcal{D}$, then with probability at least $1 - \delta$, the algorithm outputs a hypothesis $h_S \in \mathcal{H}$ satisfying:*

$$L_{\mathcal{D}}(h_S) \leq \inf_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + \epsilon.$$

1.2 VC Dimension

A key question in SLT is how expressive is our hypothesis class $\mathcal{H}$? If $\mathcal{H}$ is too expressive, it can fit arbitrary labelings of finite samples, leading to overfitting. This motivates the following:

Definition 1.3 (Shattering). *A finite set $C = \{c_1, \dots, c_m\} \subseteq X$ is said to be shattered by $\mathcal{H}$ if for every subset $S \subseteq C$, there exists a hypothesis $h \in \mathcal{H}$ such that for all $i \in \{1, \dots, m\}$,*

$$h(c_i) = 1 \iff c_i \in S.$$

Definition 1.4 (VC Dimension). *The Vapnik-Chervonenkis (VC) dimension of $\mathcal{H}$, denoted $\text{VC}(\mathcal{H})$, is the maximal size of a set $C \subseteq X$ that can be shattered by $\mathcal{H}$. If $\mathcal{H}$ can shatter sets of arbitrarily large size, we say $\text{VC}(\mathcal{H}) = \infty$.*

Theorem 1.5 (Fundamental Theorem of Statistical Learning). *A hypothesis class $\mathcal{H}$ is PAC learnable if and only if $\text{VC}(\mathcal{H}) < \infty$.*

The proof of this theorem can be found in Section 6 of [1].

Definition 1.6. *Let $\varphi(x; y) := \varphi(x_1, \dots, x_m; y_1, \dots, y_k)$ be a formula and let $\mathcal{M}$ be a model. We say that φ has the independence property with respect to $\mathcal{M}$ if for every $n \in \mathbb{N}$, there exists a sequence $\{b_i\}_{i \in n} \subseteq M^k$ such that for every subset $w \subseteq n$, there exists $a_w \in M^m$ with*

$$\mathcal{M} \models \varphi(a_w, b_i) \quad \text{if and only if} \quad i \in w.$$

Definition 1.7. *If φ does not have the independence property, we say that φ has the Not the Independence Property (NIP).*

Remark 1.8. *$\varphi(x; y)$ has IP (resp. NIP) if and only if $\varphi(y; x)$ has IP (resp. NIP)*

2 Main Result

We are now ready to state the connection of interest.

Theorem 2.1. *Let $\varphi(x; y)$ be a formula in a model $\mathcal{M}$, and let*

$$\mathcal{F}_\varphi := \{\{a \in M^{|x|} : \mathcal{M} \models \varphi(a; b)\} : b \in M^{|y|}\}.$$

Then φ has NIP if and only if $\mathcal{F}_\varphi$ has finite VC dimension.

Corollary 2.2.

3 Applications

The rest of this report will be investigating several consequences of the connection developed in section 3.

Definition 3.1. *A model $(M, <, \dots)$ is called o-minimal if for every definable set $X \subseteq M$ there exist finitely many open intervals $I_1, \dots, I_r \subseteq M \cup \{\pm\infty\}$ and a finite set $X_0 \subseteq M$ such that*

$$X = X_0 \cup I_1 \cup \dots \cup I_r.$$

Theorem 3.2. *Let T be an o-minimal theory. Then T has NIP.*

Theorem 3.3 (Wilkie, 1996). *The model*

$$\mathbb{R}_{\text{exp}} = (\mathbb{R}, <, +, \cdot, 0, 1, \exp),$$

where $\exp : \mathbb{R} \rightarrow \mathbb{R}_{>0}$ is the real exponential function, is o-minimal.

The proof of this theorem is deep and can be found in [?].

We now apply these model-theoretic results to the theory of Deep Learning. While neural networks have seen massive empirical success, ranging from the breakthrough of AlexNet [?] to modern Transformers [?], theoretical understanding of their generalization properties is still evolving. By viewing neural networks through the lens of definability in o-minimal structures, we attempt to derive rigorous learnability guarantees.

Definition 3.4. *A feedforward neural network with activation function $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ given by the composition of affine transformations and σ . Formally, the class of functions realizable by a fixed architecture with L layers is:*

$$f(x; \theta) = W_L \cdot \sigma(W_{L-1} \cdot \sigma(\dots \sigma(W_1 x + b_1) \dots) + b_{L-1}) + b_L$$

where $\theta = (W_1, \dots, W_L, b_1, \dots, b_L)$ represents the trainable weights and biases.

Since standard PAC learning (Definition 1.1) applies to binary classification, we consider the hypothesis class obtained by thresholding the network's output.

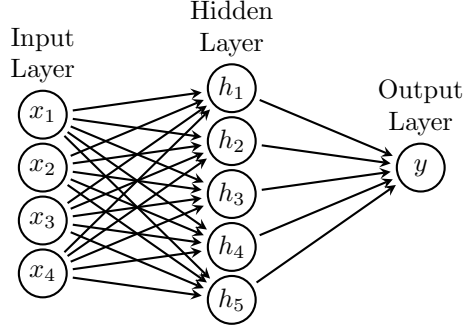


Figure 1: A simple feed-forward neural network.

Definition 3.5 (Neural Hypothesis Class). *Let $\mathcal{N}$ be a fixed neural network architecture with scalar output. We define the associated hypothesis class $\mathcal{H}_{\mathcal{N}} \subseteq \{0, 1\}^{\mathbb{R}^n}$ as:*

$$\mathcal{H}_{\mathcal{N}} := \{x \mapsto \mathbb{I}(f(x; \theta) > 0) \mid \theta \in \mathbb{R}^k\}$$

where $\mathbb{I}$ is the indicator function and k is the total number of parameters.

We can now state a result linking Deep Learning to Model Theory.

Theorem 3.6. *Let $\mathcal{H}_{\mathcal{N}}$ be a neural network hypothesis class where the activation function σ is definable in $\mathbb{R}_{\text{exp}}$ (e.g., Sigmoid, Tanh, or ReLU). Then $\mathcal{H}_{\mathcal{N}}$ is PAC learnable.*

Proof. The function $f(x; \theta)$ is a composition of the activation function σ and affine operations $(+, \times)$, which are all definable in $\mathbb{R}_{\text{exp}}$. Thus, the formula $\varphi(x; \theta) := "f(x; \theta) > 0"$ is definable in $\mathbb{R}_{\text{exp}}$. The definable family associated with this formula, $\mathcal{F}_{\varphi} = \{\{x \in \mathbb{R}^n : \mathcal{M} \models \varphi(x; \theta)\} : \theta \in \mathbb{R}^k\}$, coincides exactly with the family of support sets for the neural hypothesis class $\mathcal{H}_{\mathcal{N}}$. Since $\mathbb{R}_{\text{exp}}$ is o-minimal (Theorem 4.3), its theory has NIP (Theorem 4.2), implying that every formula definable in $\mathbb{R}_{\text{exp}}$, including our network formula $\varphi(x; \theta)$, has NIP. Therefore, by Theorem 3.1, the family $\mathcal{F}_{\varphi}$ has finite VC dimension. As $\text{VC}(\mathcal{H}_{\mathcal{N}}) = \text{VC}(\mathcal{F}_{\varphi}) < \infty$, it follows from the Fundamental Theorem of Statistical Learning (Theorem 1.5) that $\mathcal{H}_{\mathcal{N}}$ is PAC learnable. $\square$

3.1 Rado Graph

To fully appreciate the significance of NIP it is instructive to examine a structure where NIP fails completely. We illustrate the classical example of such a structure below.

Definition 3.7. *The Rado Graph $\mathcal{R} = (V, E)$ is the unique (up to isomorphism) countable graph satisfying the Extension Property: For any two disjoint finite sets of vertices $U, V \subset \mathcal{R}$, there exists a vertex $z \in \mathcal{R} \setminus (U \cup V)$ such that:*

- z is connected to every vertex in U (i.e., $\forall u \in U, E(z, u)$),

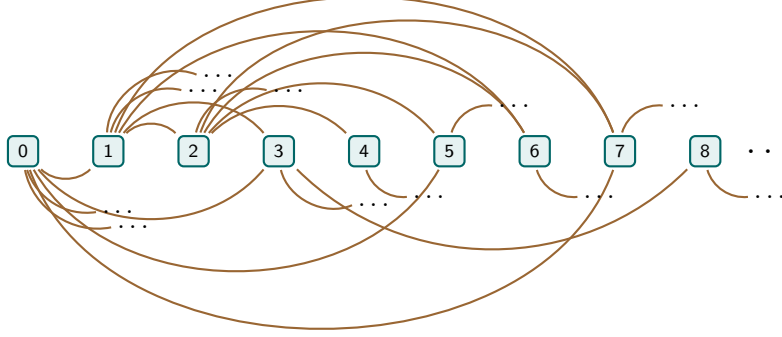


Figure 2: Rado graph

- z is connected to no vertex in V (i.e., $\forall v \in V, \neg E(z, v)$).

Intuitively, this graph can be constructed probabilistically by taking countably many vertices and connecting each pair with probability $1/2$. The resulting structure is "random" enough that it contains every possible finite graph as an induced subgraph.

Theorem 3.8. *The theory of the Rado Graph has the Independence Property (IP).*

Proof. Let $\varphi(x; y) := E(x, y)$ be the edge relation, and $A = \{a_1, \dots, a_n\}$ be any finite set of vertices in $\mathcal{R}$. We must show that for *every* subset $S \subseteq A$, there exists a parameter b_S such that $\varphi(b_S; a_i)$ holds if and only if $a_i \in S$.

To that end, let $U = S$ and $V = A \setminus S$. The Extension Property guarantees the existence of a vertex z such that z is connected to everything in U and nothing in V .

Setting $b_S = z$, we see that the formula $E(x, y)$ can encode every possible binary pattern on A . Since this holds for any finite n , the VC dimension is infinite. Thus, the theory has IP. $\square$

4 Discussion

Our work has been limited to binary outputs, but one can easily extend to allow for other sorts of output by considering various different types of loss functions (see [1, Section 3.2]).

In section four we showed that o-minimality and NIP provide theoretical guarantees that certain neural network architectures are PAC learnable. Unfortunately these results do not fully capture the generalization behavior observed in modern deep networks. Zhang et al. [4] experimentally demonstrated that

large neural networks can perfectly fit random labels and even unstructured noise, indicating that finite VC-dimension or PAC-learnability alone does not explain why deep networks generalize well in practice.

5 Conclusion

References

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